

Wubing (Ryan) Xia

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EDUCATION

University of Illinois at Urbana-Champaign (UIUC), Champaign, IL Aug 2023–Jun 2027

BS in Computer Engineering, GPA: 3.70/4 Research Advisor: [Prof. Lingming Zhang](#)

Relevant Coursework: Artificial Intelligence, Parallel Programming, Database Systems, Computer Systems Engineering, Digital Systems Laboratory

Zhejiang University, Zhejiang, China Aug 2023–Jun 2027

BEng in Electronic and Computer Engineering, GPA: 3.97/4.3

Relevant Coursework: Data Structures, Computer Systems & Programming, Linear Algebra, Discrete Math, Probability with Engineering Applications

RESEARCH EXPERIENCE

Recursive Self-Improvement of Agent Harnesses: Where Do the Gains Come From? Jun 2026–Present

Research Assistant, Zhangyang (Atlas) Wang, UT. Austin; paper under review at ICLR 2027

- Built a Harbor-based evaluation framework for harness self-improvement methods (Meta-Harness, AHE, Retro Harness) on Terminal-Bench 2.1, DeepSWE, and FrontierCS, with held-out task splits and repeated-run statistics.
- Ran each method's evolution on each benchmark. For each evolved harness and its seed 8 times per held-out task (~3,200 runs). We found only **2 of 8** sessions improved reliably, 5 were indistinguishable from noise, and 1 reliably regressed.
- Showed that neither reliable gain, both from Meta-Harness, reflects a more capable agent: on Terminal-Bench (+14.5 pp), the evolved harness recovers crashed shells; on FrontierCS (+5.0 pp), it spends more test-time compute on an end-of-run self-check.

Can Frontier LLM Agents Write PoCs for Real-World Vulnerabilities? Jan 2026–Jun 2026

Research Assistant, Lingming Zhang's Lab, UIUC; SEC-bench Pro, under review at ICSE 2027

Used by OpenAI in the full evaluations of [GPT-5.5-Cyber](#) and [GPT-5.6](#).

- Contributed to constructing SEC-bench Pro, a benchmark for evaluating LLM agents' ability to perform real-world vulnerability analysis and PoC (proof-of-concept) reproduction on complex software systems.
- Expanded the benchmark to improve coverage of security scenarios by adding reproducible vulnerability tasks from V8, Linux, and SpiderMonkey, including **100+** additional tasks, and integrating syzbot as additional sources of Linux instances.
- Designed and implemented an LLM-agent-driven benchmark generation pipeline to automate vulnerability environment reconstruction, PoC reproduction, and validation; totally generated **56** validated vulnerability tasks.
- Developed fix-verify validation and privilege audit mechanisms to identify failed reproductions and prevent evaluation bias caused by incorrect privilege assumptions.

A "Spot the Difference" Challenge for Large Multimodal Models Sep 2025–Dec 2025

Research Assistant, CVNext Lab; Supervisor: Prof. Gaoang Wang

- Built an automated evaluation pipeline using OpenAI and Gemini APIs to benchmark multimodal LLMs' reasoning capabilities, conducting evaluations on **2,932** QA pairs from **270** simulation environments and **552** QA pairs from **29** physical environments.
- Resolved video input limitations across LLM APIs by developing a video-to-image pipeline and key-frame sampling strategy to enable GPT-5 video evaluation and overcome Gemini-2.5's video length limits.
- Designed query scheduling strategies by controlling request frequency and introducing execution intervals between batches to ensure reliable large-scale LLM evaluation.

PUBLICATION

1. Jiajun Zhu, **Wubing Xia**, Xu Yang, Jingzhe Liu, Lingming Zhang, Zhangyang Wang, Tong Zhao, Neil Shah, Mingxuan Ju. *Can Harness Optimization Improve Agents? Generalizable Gains Versus Evaluation Noise* **ICLR 2027 (Submission)**
2. Hwiwon Lee, Jiawei Liu, Dongjun Kim, **Wubing Xia**, Ziqi Zhang, Chunqiu Steven Xia, Lingming Zhang. *SEC-bench Pro: Can Language Models Solve Long-Horizon Software Security Task?* **ICSE 2027 (Submission)**
3. K.W. Wei, B.C. Hu, J. Cao, X.H. Chen, Z.X. Lu, **W.B. Xia**, W.L. Xu, J.A. Wu, J.C. He, M.Y. Jia, C.Y. Zhao, Y. Sun, Y.Z. Li, Z.H. Zhao, J. Zhang, G.A. Wang. *M³-Verse: A "Spot the Difference" Challenge for Large Multimodal Models*. **Manuscript**

COURSEWORK

1. GPT-2 CUDA Kernel Optimizations
2. RTL-based FPGA Snake Game Development
3. UNIX-like Operating System Kernel using C and RISC-V Assembly

ADDITIONAL

Competition Awards: CCF-NOIP2022 (First Prize), Codeforces rating: 2211(master)

Skills: C, C++, Python, CUDA, FPGA, System Verilog

Languages: English (fluent), Mandarin (native)

Table Tennis Awards

- China: 2024 ZJU "San Hao Cup" Men's Singles Champion
- USA: 2025-2026 Season NCTTA (National Coded Team 11th, Regional Coded Team 2nd, Divisional Men's Single 4th)